Equations

Energy is $E = m c^2$ and the golden ratio is $\varphi = \frac{1+\sqrt{5}}{2}$.

The sample mean is $\acute{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$ for data $x_1, \ldots, x_n$.

$$\int_0^{\infty} e^{-x^2}\, d\,x = \frac{\sqrt{\pi}}{2}$$

Bayes: $p(\theta / x) = \frac{p(x / \theta)\, p(\theta)}{p(x)}$.

$$\nabla \cdot E = \frac{\rho}{\varepsilon_0}$$

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, det\ A = a\, d - b\, c$$

$$f(x) = \begin{cases} x^2 & x \geq 0 \\ -x & x < 0 \end{cases}$$

$$\lim_{n \to \infty} \left(1 + \frac{1}{n}\right)^n = e$$

The norm $\left\| v \right\|_2 = \sqrt{\sum_k v_k^2}$ and $\hat{\beta} = (X^T X)^{-1} X^T y$.

$$\sum_{k=0}^{n} \binom{n}{k} = 2^n, \prod_{j=1}^{m} a_j \leq \max_j a_j^m$$